Special Cases

(i) For a purely resistive circuit, the phase angle between voltage and current is zero and cos $\phi = 1$.

$$\therefore P_{av} = V_{RMS}\ I_{RMS}$$

(ii) For a purely inductive or capacitive circuit, the phase angle is $\pm \pi/2$ and $\cos\left(\pm \pi/2\right) = 0$.

$$\therefore P_{av} = 0$$

(iii) For series RLC circuit, the phase angle $\phi = \tan^{-1}\left(\dfrac{X_L - X_C}{R}\right)$

$$\therefore P_{av} = V_{RMS}\ I_{RMS}\ \cos\phi$$

(iv) For series RLC circuit at resonance, the phase angle is zero and $\cos\phi = 1$.

$$\therefore P_{av} = V_{RMS}\ I_{RMS}$$

4.8.2 Wattless current

Consider an AC circuit in which there is a phase angle of ϕ between V_{RMS} and I_{RMS} and voltage is assumed to be leading the current by ϕ as shown in the phasor diagram (Figure 4.50).

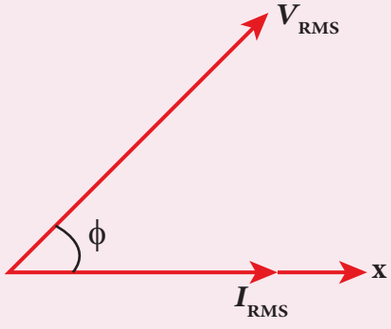


Figure 4.50 V_{RMS} leads I_{RMS} by ϕ

Now, I_{RMS} is resolved into two perpendicular components, namely $I_{RMS}\cos\phi$ along V_{RMS} and $I_{RMS}\sin\phi$ perpendicular to V_{RMS} as shown in Figure 4.51.

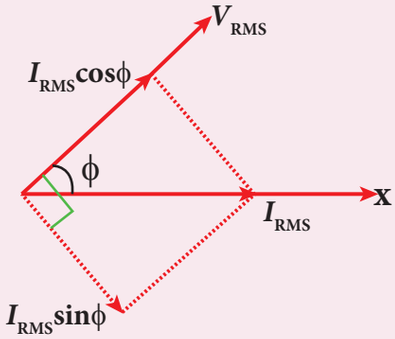


Figure 4.51 The components of I_{RMS}

(i) The component of current $\left(I_{RMS}\cos\phi\right)$ which is in phase with the voltage is called active component. The power consumed by this current $= V_{RMS} I_{RMS}\cos\phi$. So that it is also known as 'Wattful' current.

(ii) The other component $\left(I_{RMS}\sin\phi\right)$ which has a phase angle of $\pi/2$ with the voltage is called reactive component. The power consumed is zero. Hence it is also known as 'Wattless' current.

The current in an AC circuit is said to be wattless current if the power consumed by it is zero. This wattless current occurs in a purely inductive or capacitive circuit.

4.8.3 Power factor

The power factor of a circuit is defined in one of the following ways:

(i) **Power factor = cos ϕ = cosine of the angle of lead or lag**

(ii) **Power factor** $= \dfrac{R}{Z} = \dfrac{\text{Resistance}}{\text{Impedance}}$

(iii) **Power factor** $= \dfrac{P_{av}}{V_{RMS} I_{RMS}}$

$= \dfrac{\text{True power}}{\text{Apparent power}}$

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